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TUTORIAL

Frequency Response of Amplifiers

MOSFET Capacitances, Miller Effect, Association of Poles, and the Frequency Response of CS, Source-Follower, CG, Cascode, and Differential Stages

Department of Electronics and Information Engineering | Ho Chi Minh City University of Technology and Engineering (HCM-UTE), Vietnam

Based on Design of Analog CMOS Integrated Circuits, Chapter 6 (Razavi)

OUTLINE

Roadmap for This Lecture

Eight sections, from MOSFET capacitances to the differential-pair OTA

I MOSFET Capacitances & Pole Fundamentals

- Physical origin and region-dependent formulas for CGS, CGD, CGB, CDB, CSB
- Complete large-signal / small-signal MOSFET model

II The Miller Effect

- General Miller impedance-splitting theorem
- Ex. 6.1: Miller capacitance of an ideal inverter

III Association of Poles with Nodes

- Non-interacting vs. interacting poles
- Ex. 6.3: pole at node X via Miller + association

IV Common-Source Stage — Frequency Response

- Miller approximation (small RS) vs. OCTC method (large RS)
- Exact transfer function, dominant and non-dominant poles

V Source Follower — Frequency Response

- Exact transfer function and OCTC dominant pole
- Input impedance at low and high frequency

VI Common-Gate Stage — Frequency Response

- No Miller capacitance — wider bandwidth than CS
- Ex. 6.7: input impedance independent of CL

VII Cascode Stage — Frequency Response

- Pole at each node; suppressed Miller effect at the input
- Gain, input/output resistance vs. plain CS

VIII Differential Pair — Frequency Response

- Differential/common-mode half-circuits and CMRR vs. frequency
- 5-transistor current-mirror-load OTA: mirror pole, output pole, zero

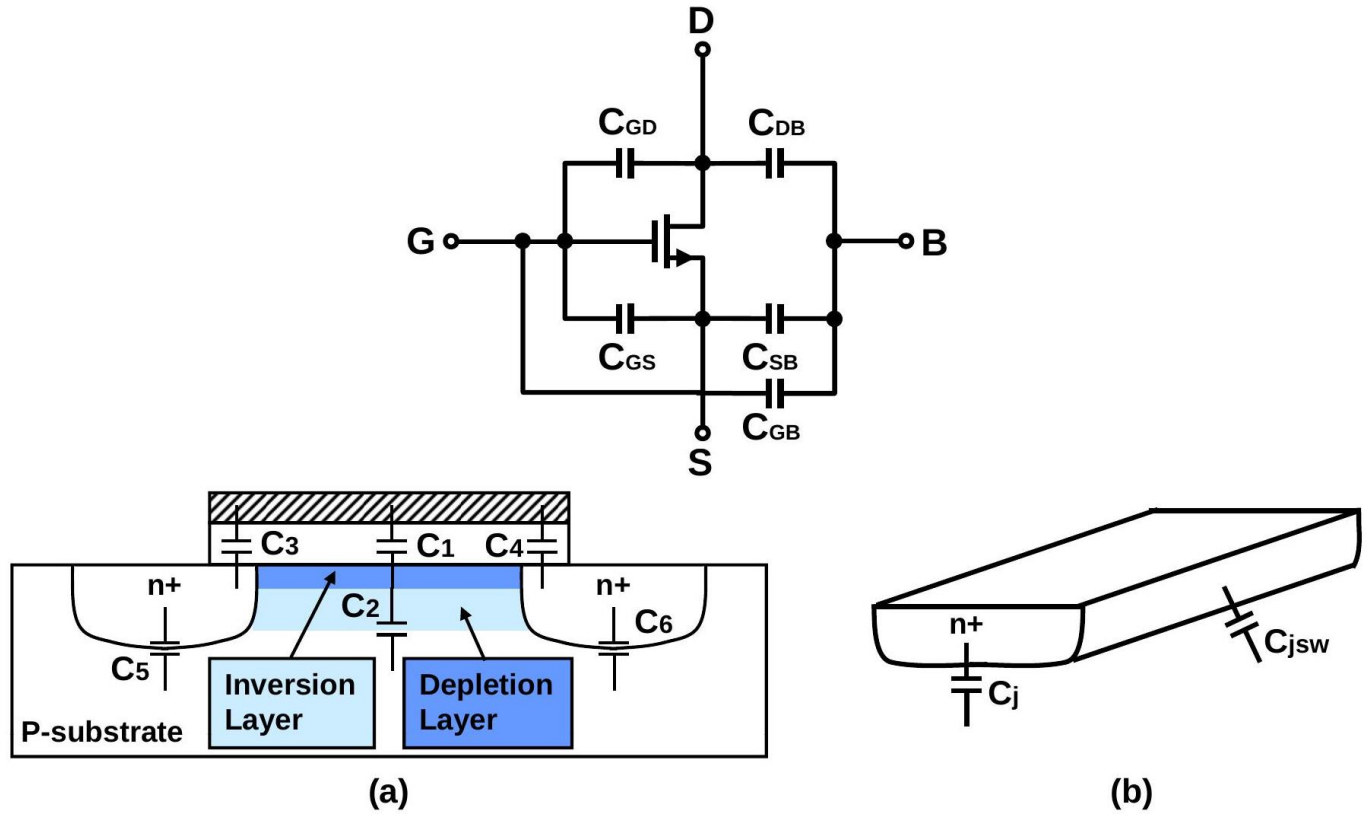


MOSFET Capacitances & Pole Fundamentals

- Physical origin and region-dependent formulas for CGS, CGD, CGB, CDB, CSB
- Complete large-signal / small-signal MOSFET model
- Review: the pole of a first-order RC circuit

MOSFET Parasitic Capacitances — Physical Origin

- Five capacitances link D, G, S, B: CGD, CDB, CGS, CSB, CGB — used in every HF small-signal model in this chapter
- Physical origin: C1 gate-channel oxide, C2 channel-substrate (depletion), C3/C4 gate overlap, C5/C6 junction
- Junction capacitance = area term Cj + sidewall term Cjsw, both voltage-dependent (reverse-biased pn junctions)



MOSFET Capacitance Formulas — Linear and Saturation

- Linear region: CGD = CGS = (1/2)W·Leff·Cox + W·Cov — channel charge splits evenly between source and drain
- Saturation: CGD = W·Cov (channel pinched off at drain); CGS = (2/3)W·Leff·Cox + W·Cov, the classic Meyer 2/3 coefficient
- Both regions: CGB ≈ CGB,overlap only; CSB = CDB = junction capacitance

➢**Linear Region**

$$C_{GD} = C_{GS} = \frac{1}{2} W L_{eff} C_{ox} + W C_{ov}$$

$$C_{GB} = C_{GB,overlap}$$

➢**Saturation Region**

$$C_{GD} = W C_{ov}$$

$$C_{GS} = \frac{2}{3} W L_{eff} C_{ox} + W C_{ov}$$

$$C_{GB} = C_{GB,overlap}$$

$$C_{SB} = C_{DB} = \text{junction capacitance}$$

MOSFET Capacitance Formulas — Off Region

- VGS < VTH, no channel: CGD = CGS = W·Cov — only overlap terms remain
- CGB = Coxide in series with Cdep, plus CGB,overlap — no channel to shield gate from bulk
- Together with slides 2-3, this completes the capacitance model needed at any bias point

➢**Off Region**

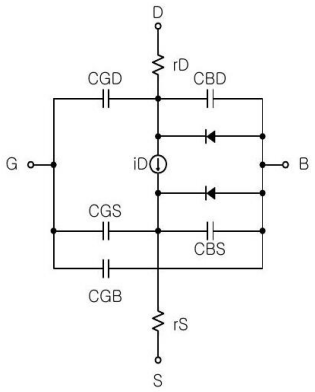
$$C_{GD} = WC_{ov}$$

$$C_{GS} = WC_{ov}$$

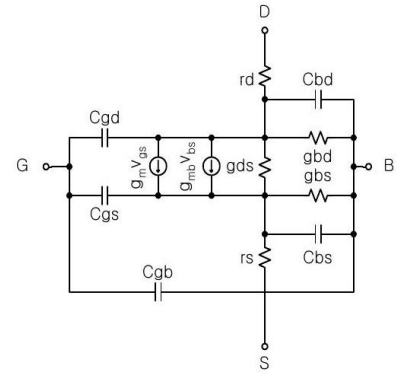
$$C_{GB} = C_{oxide} \parallel C_{dep} + C_{GB,overlap}$$

Complete Large-Signal and Small-Signal MOSFET Model

- Large-signal (left): 4-terminal device, iD, rD/rS, junction diodes with CBD/CSB, plus CGD/CGS/CGB; Shichman-Hodges I-V equations
- Small-signal (right): dependent sources gm·vgs and gmb·vbs (body effect), gds = 1/ro, same capacitor set
- id = gm·vgs + gd·vds − gmb·vsb — the master model for every stage analyzed in this chapter



Large−signal equivalent circuit



Small−signal equivalent circuit

$$i_d = g_m v_{gs} + g_d v_{ds} - g_{mb} v_{sb}$$

$$I = \frac{\mu C_{ox}}{2}\left(\frac{W}{L}\right)\left\{2(V_{GS} - V_T) - \frac{V_{DS}}{2}\right\}V_{DS}$$

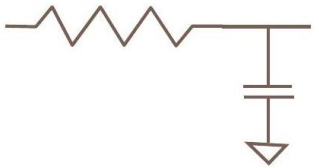
$$I = \frac{\mu C_{ox}}{2}\left(\frac{W}{L}\right)(V_{GS} - V_T)^2(1 + \lambda V_{DS})$$

Review — the Pole of a First-Order RC Circuit

- vi through R into node vo with C to ground: vo/vi = 1/(1+sRC) = 1/(1+j·f/fp), pole frequency fp = 1/(2πRC)
- At f = fp, magnitude drops to 1/√2 (−3 dB), phase lags 45°; well above fp, rolloff is 20 dB/decade
- This R·C rule at a node is the basic tool behind 'association of poles with nodes' later in the chapter

$$vo/vi = \frac{1/sC}{R+1/sC}$$

$$vo/vi = \frac{1}{sRC+1}$$



$$\frac{vo}{vi}(f) = \frac{1}{1+j2\pi fRC}$$

$$\frac{vo}{vi}(f) = \frac{1}{1+j\,f/fp}, \quad fp = 1/2\pi RC$$

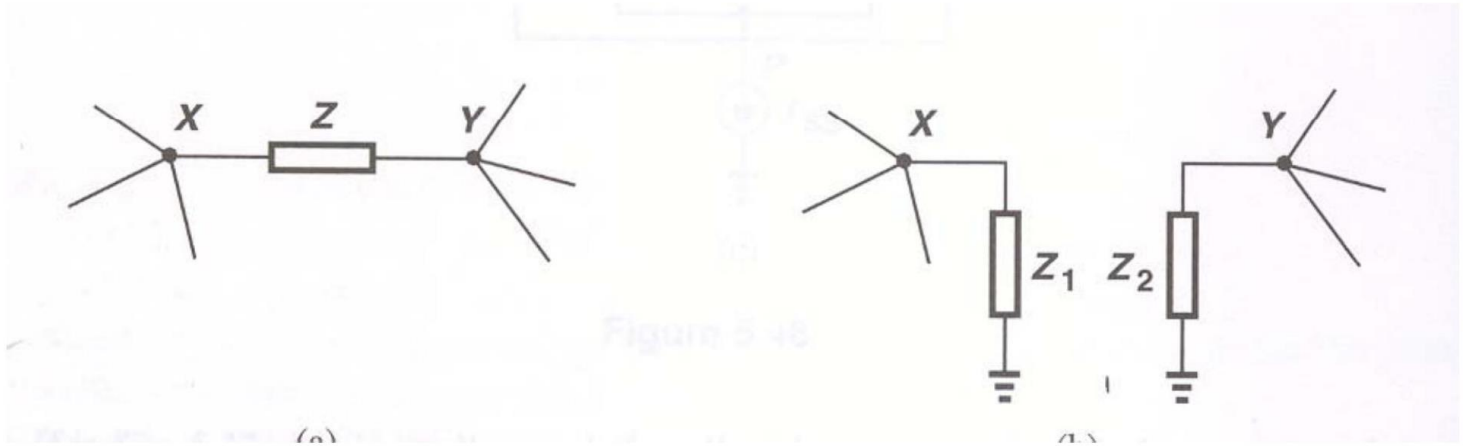
II

The Miller Effect

- General Miller impedance-splitting theorem
- Ex. 6.1: Miller capacitance of an ideal inverter
- Ex. 6.2: Miller transform applied to a resistor

The Miller Effect — General Derivation

- Impedance Z between nodes X, Y with gain Av = Vy/Vx splits into grounded Z1 (at X) and Z2 (at Y) drawing the same currents
- Z1 = Z/(1−Av); Z2 = Z/(1−1/Av)
- For large negative Av (high-gain inverter), Z1 shrinks sharply — a small feedback C looks like C·(1−Av) at the input: the Miller multiplication



$$I_X = \frac{V_X - V_Y}{Z} = \frac{V_X - A_v V_X}{Z} = \frac{1 - A_v}{Z} V_X$$

$$\therefore Z_1 = \frac{Z}{(1 - A_v)}$$

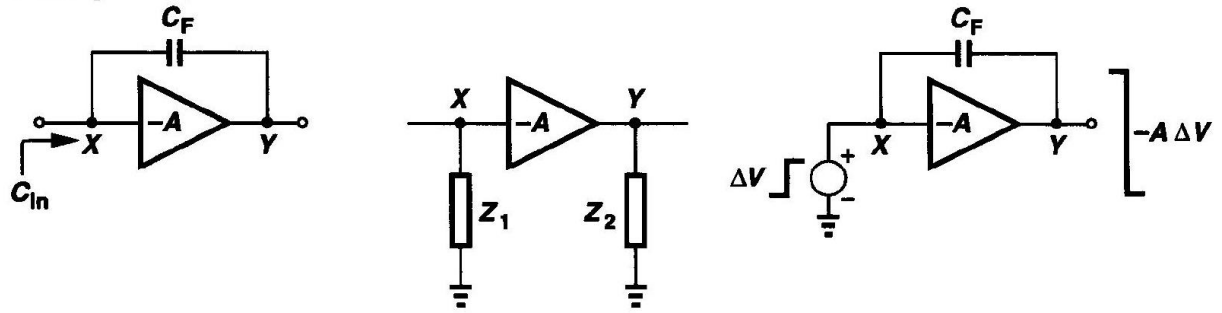
$$I_Y = \frac{V_Y - V_X}{Z} = \frac{V_Y - \frac{V_Y}{A_v}}{Z} = \frac{V_Y}{Z/(1 - 1/A_V)}$$

$$\therefore Z_2 = \frac{Z}{(1 - A^{-1}{}_v)}$$

Ex. 6.1 — Miller Capacitance of an Ideal Inverter

- Ideal inverting amp, gain −A, feedback CF between input X and output Y
- Input-side: C1 = CF·(1−Av) = CF·(1+A) — CF appears (1+A)× larger at the input
- Output-side: C2 = CF·(1−1/Av) ≈ CF — essentially unchanged at the output node

Ex6.1)



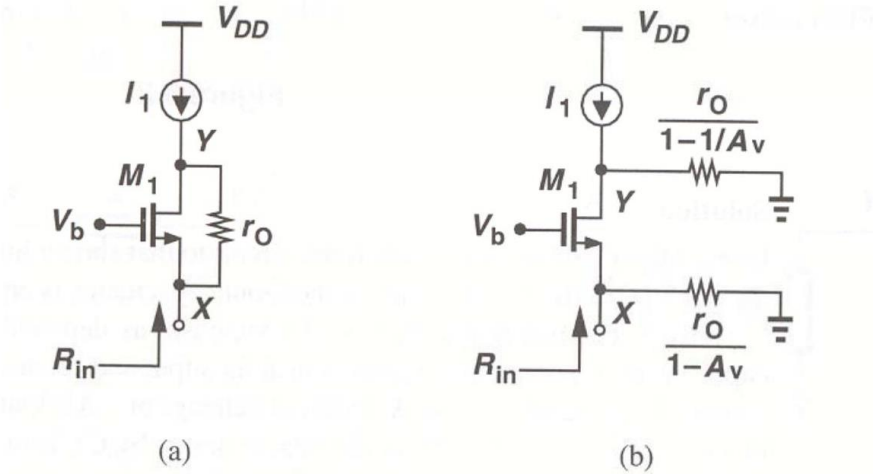
$$C_1 = C_F(1 - A_v)$$

$$C_2 = C_F(1 - A^{-1}_v) \approx C_F$$

Ex. 6.2 — Miller Transform Applied to a Resistor

- M1: gate at fixed Vb, source at node X (=Rin), drain at Y with ro to ground; Av = Vy/Vx = 1+(gm+gmb)ro
- Miller Z1 = ro/(1−Av) = −1/(gm+gmb) — a negative resistance appears at X
- This cancels the source's intrinsic +1/(gm+gmb) exactly: Rin → ∞

Example 6.2)



$$A_v = 1+(g_m+g_{mb})r_o$$

$$\frac{r_o}{1-A_v} = \frac{r_o}{1-[1+(g_m+g_{mb})r_o]} = -\frac{1}{g_m+g_{mb}}$$

$$\therefore R_{in} = \frac{1}{g_m+g_{mb}} \| \left(-\frac{1}{g_m+g_{mb}}\right) = \infty$$



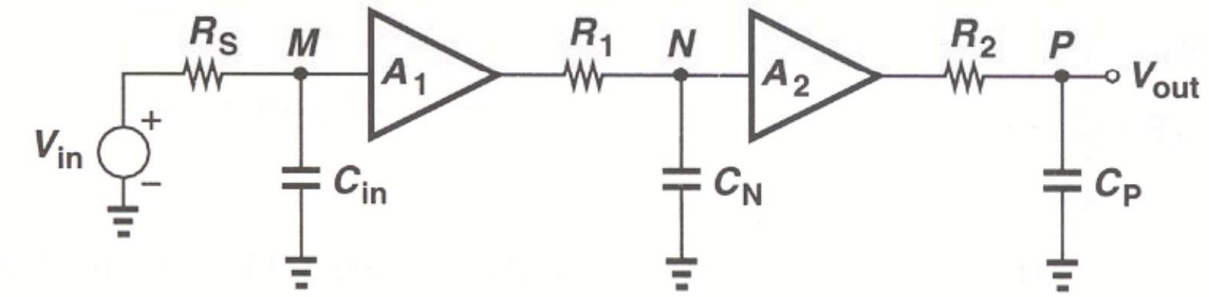
Association of Poles with Nodes

- Non-interacting vs. interacting poles
- Ex. 6.3: pole at node X via Miller + association
- Ex. 6.4: two-node Gm/Rout derivation

Association of Poles with Nodes — Non-Interacting

- Cascade Vin → RS/Cin(M) → A1 → R1/CN(N) → A2 → R2/CP(P) → Vout, with A1, A2 unilateral (no feedback)
- Each node behaves as an independent first-order RC low-pass since signal only flows forward
- Vout/Vin = A1A2 · 1/(1+RSCins)·1/(1+R1CNs)·1/(1+R2CPs) — one pole per node, using R and C at that node alone

➢Non-Interacting Poles

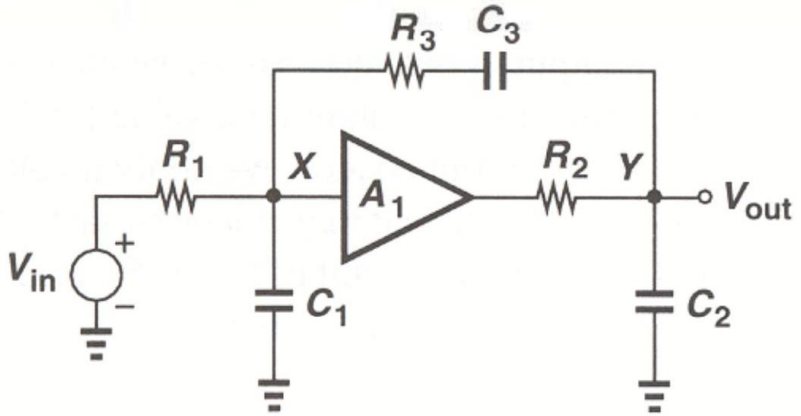


$$\therefore \frac{V_{out}}{V_{in}} = A_1 A_2 \frac{1}{1+R_S C_{in} s} \cdot \frac{1}{1+R_1 C_N s} \cdot \frac{1}{1+R_2 C_P s}$$

Association of Poles with Nodes — Interacting

- Same cascade, but feedback branch R3+C3 now connects output node Y directly back to input node X
- This couples the node equations at X and Y — they can no longer be solved independently
- Result: a true second-order denominator whose poles depend jointly on R3, C3, and gain A1 — not simply 1/(R1C1), 1/(R2C2)

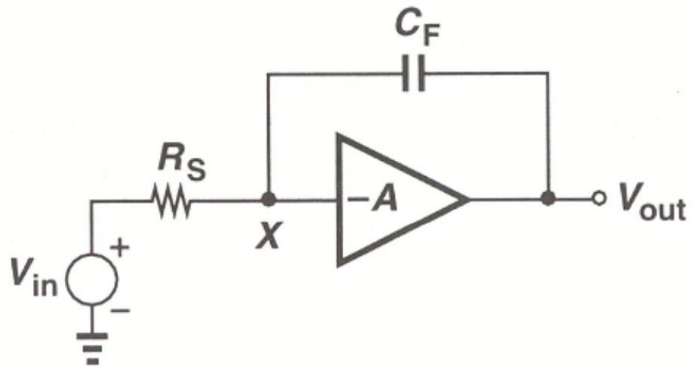
➢**Interacting Poles**



Ex. 6.3 — Pole at Node X via Miller + Association

- Inverting amp (gain −A), source resistance RS, feedback CF between input X and output
- Miller-transform CF to (1+A)·CF at X, reducing node X to a simple RC circuit with RS
- Pole frequency = 1/[RS(1+A)CF] rad/s — small CF with large loop gain A gives a very low dominant pole (dominant-pole compensation)

Example 6.3) calculate the pole associated with node X

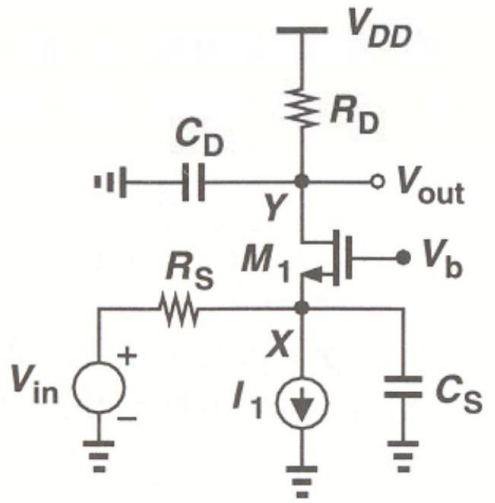


$$pole\ \ frequency = 1/[R_S(1+A)C_F]\left(rad/\sec\right)$$

Ex. 6.4 (Setup) — Two-Node Model with CS and CD

- M1 with source node X (bias I1, capacitance CS) and drain node Y (load RD, capacitance CD to Vout)
- Input pole: ωin = {(CGS1+CSB1)·[RS ‖ 1/(gm1+gmb1)]}$^{-1}$
- Output pole: ωout = {(CGD1+CDB1)·RD}$^{-1}$ — one pole per node, using actual MOSFET R and C values

Example 6.4)



$$\omega_{in} = \left\{(C_{GS1} + C_{SB1}) \cdot \left[R_S \parallel \frac{1}{g_{m1} + g_{mb1}}\right]\right\}^{-1}$$

$$\omega_{out} = \{(C_{GD1} + C_{DB1}) \cdot R_D\}^{-1}$$

Ex. 6.4 — Complete Two-Pole Transfer Function

- Vout/Vin = Gm·Rout · 1/[(1+s/ωin)(1+s/ωout)] — DC gain Gm·Rout times two independent node poles
- Gm = (gm1+gmb1)/[1+(gm1+gmb1)RS] — source resistance RS degenerates the effective transconductance
- Gm and Rout are each derived independently on the next two slides to verify this result

Example 6.4)

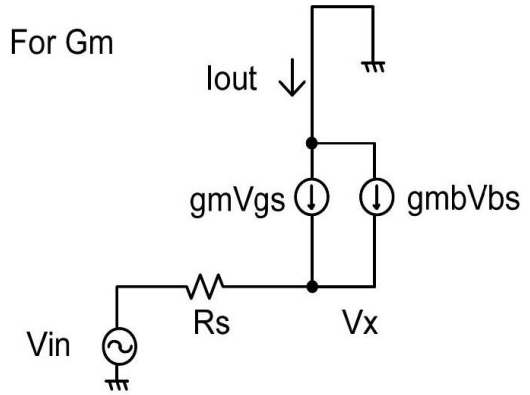
$$\therefore V_{out}/V_{in} = G_m R_{out} \frac{1}{(1+\frac{s}{\omega_{in}})(1+\frac{s}{\omega_{out}})}$$

$$= \frac{g_{m1}+g_{mb1}}{1+(g_{m1}+g_{mb1})R_S} R_D \frac{1}{(1+\frac{s}{\omega_{in}})(1+\frac{s}{\omega_{out}})}$$

Ex. 6.4 — Deriving Gm from the Small-Signal Circuit

- Gate and bulk at signal ground ⟹ Vgs = Vbs = −VX, so gm(−VX)+gmb(−VX) = iout
- Loop equation: iout·RS + Vin = VX
- Solving: Gm = iout/Vin = −(gm+gmb)/[1+(gm+gmb)RS] — matches slide 14's magnitude

Example 6.4)



$$i_{out}R_S + V_{in} = V_X$$

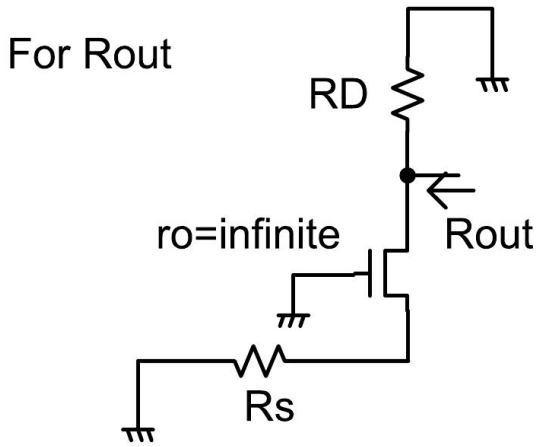
$$g_m(-V_X) + g_{mb}(-V_X) = i_{out}$$

$$\therefore G_m = \frac{i_{out}}{V_{in}} = -\frac{g_m + g_{mb}}{1+(g_m + g_{mb})R_S}$$

Ex. 6.4 — Deriving Rout (Assuming ro = ∞)

- With ro infinite (channel-length modulation neglected), the transistor adds no finite path from drain to ground
- Output resistance is set purely by the load: Rout = RD
- Combined with Gm, this completes Gm·Rout for the transfer function of slide 14

Example 6.4)



$$\therefore R_{out} = R_D$$

IV

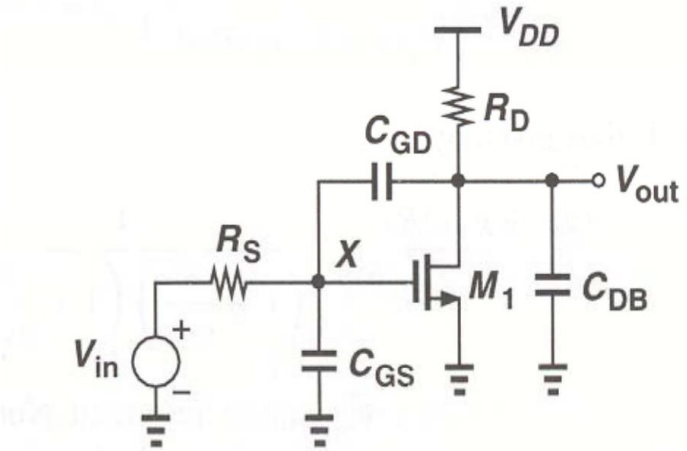
Common-Source Stage — Frequency Response

- Miller approximation (small RS) vs. OCTC method (large RS)
- Exact transfer function, dominant and non-dominant poles
- Right-half-plane zero and input impedance

Common-Source Stage — Miller Approximation (RS Small)

- Standard CS stage: RS to gate X (CGS to ground), feedback CGD to drain output, RD load, CDB at output
- Input pole: ωp,in = 1/{RS·[CGS+(1+gmRD)·CGD]} — CGD Miller-multiplied by (1+gmRD)
- Output pole: ωp,out = 1/[(CGD+CDB)·RD] — input pole is usually dominant

when Rs is small,



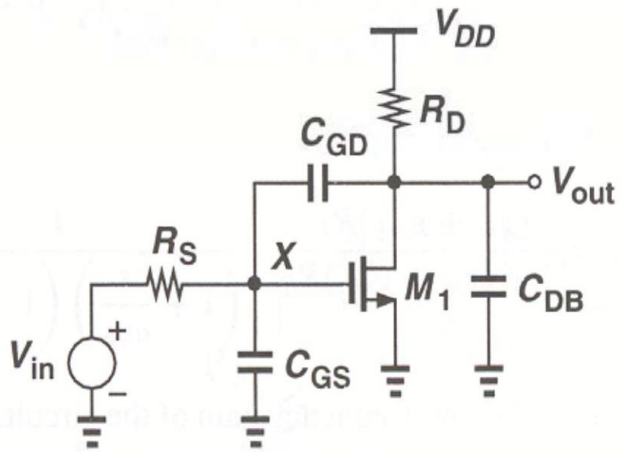
$$\omega_{p,in} = \frac{1}{R_S\left[C_{GS} + (1 + g_m R_D)C_{GD}\right]}$$

$$\omega_{p,out} = \frac{1}{\left[(C_{GD} + C_{DB})R_D\right]}$$

Common-Source Stage — OCTC Method (RS Large)

- When RS is large, the simple one-directional Miller approximation for CGD breaks down
- Effective C = (CGD‖CGS)+CDB; effective R = RD ‖ {1/[gm·CGD/(CGS+CGD)]}
- τ = RC, ω = 1/τ — the slide-17 Miller formula is just the RS-small special case of this general result

➢**when Rs is very large**



$$C = (C_{GD} \| C_{GS}) + C_{DB}$$

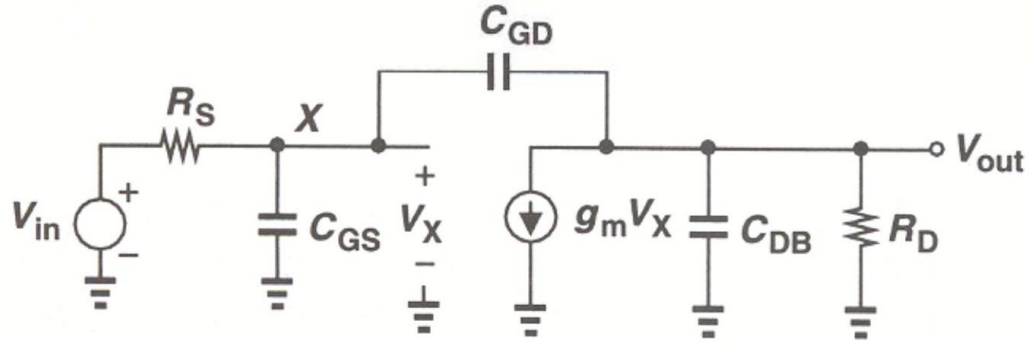
$$R = R_D \| \frac{1}{g_m \frac{C_{GD}}{C_{GS}+C_{GD}}}$$

$$\because \tau = RC = \left((C_{GD} \| C_{GS}) + C_{DB}\right)\left(R_D \| \frac{1}{g_m \frac{C_{GD}}{C_{GS}+C_{GD}}} \right)$$

$$\because \omega = 1/\tau$$

Common-Source Stage — Exact Transfer Function

- Full two-node KCL (no Miller/OCTC approximation): CGS at X, CGD feeding back to output, gm·VX and CDB/RD at output
- vo/vi = (sCGD−gm)RD / {s²RSRD(CGSCGD+CGSCSB+CGDCDB) + s[RS(1+gmRD)CGD+RSCGS+RD(CGD+CDB)]+1}
- At s→0: DC gain −gmRD; as s grows, sCGD can cancel gm, producing a zero (analyzed later)



$$\frac{vo}{vi} = \frac{(sC_{GD} - g_m)R_D}{s^2 R_S R_D (C_{GS}C_{GD} + C_{GS}C_{SB} + C_{GD}C_{DB}) + s\left[R_S(1+g_m R_D)C_{GD} + R_S C_{GS} + R_D(C_{GD} + C_{DB})\right] + 1}$$

OCTC Approximation — Extracting the Dominant Pole

- D = (s/ωp1+1)(s/ωp2+1); if ωp2 >> ωp1, D ≈ s²/(ωp1ωp2) + s/ωp1 + 1
- Matching the linear-in-s coefficient to the exact denominator (slide 19) reads off the dominant pole directly
- fp,in = 1/{2π·(RS[CGS+(1+gmRD)CGD] + RD(CGD+CDB))} — note the extra RD(CGD+CDB) term vs. the simple Miller formula

$$Assume\ D = \left(\frac{s}{\omega_{p1}}+1\right)\left(\frac{s}{\omega_{p2}}+1\right) = \frac{s^2}{\omega_{p1}\omega_{p2}} + \left(\frac{s}{\omega_{p1}} + \frac{s}{\omega_{p2}}\right) + 1,$$

$$\omega_{p2} >> \omega_{p1}, D \approx \frac{s^2}{\omega_{p1}\omega_{p2}} + \left(\frac{s}{\omega_{p1}}\right) + 1$$

$$f_{p,in} = \frac{1}{2\pi\left(R_S\left[C_{GS} + (1+g_m R_D)C_{GD}\right] + R_D(C_{GD}+C_{DB})\right)}$$

OCTC Approximation — the Non-Dominant Pole fp,out

- Matching the s^2 coefficient (using ω_{p1} already found) yields fp,out
- Large CGS limit: fp,out ≈ 1/[2πRD(CGD+CDB)] — recovers the simple Miller output pole
- Large CGD limit: fp,out ≈ gm/[2π(CGS+CDB)] — the second pole is set by gm, not RD

$$f_{p,out} = \frac{R_S(1+g_m R_D)C_{GD} + R_S C_{GS} + R_D(C_{GD}+C_{DB})}{2\pi R_S R_D(C_{GS}C_{GD} + C_{GS}C_{DB} + C_{GD}C_{DB})}$$

$$f_{p,out} \approx \frac{1}{2\pi R_D(C_{GD}+C_{DB})}, \quad \text{for large } C_{GS}$$

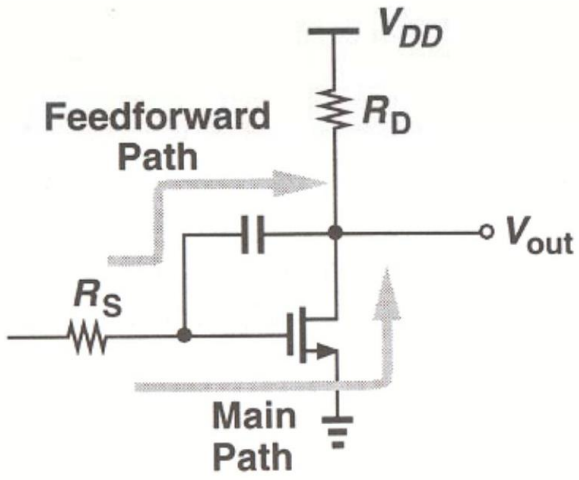
$$f_{p,out} \approx \frac{g_m R_S R_D C_{GD}}{2\pi R_S R_D(C_{GS}C_{GD} + C_{GS}C_{DB} + C_{GD}C_{DB})}$$

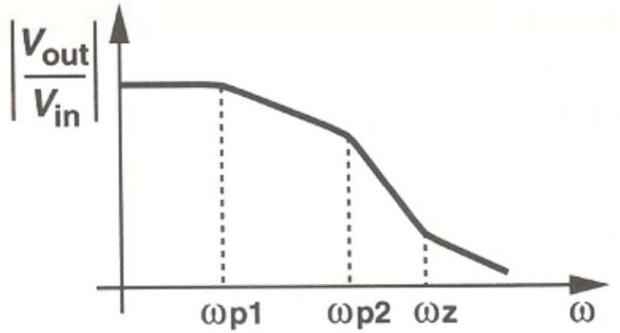
$$\approx \frac{gm}{2\pi(C_{GS}+C_{DB})}, \quad \text{for large } C_{GD}$$

Common-Source Zero — Feedforward Path Through CGD

- Two parallel paths gate→drain: Main Path through the transistor (inverting) and Feedforward Path through CGD (non-inverting)
- Bode plot: three breakpoints — ωp1 (dominant), ωp2 (second pole, −40 dB/dec), ωz (zero, flattens to −20 dB/dec)
- At high frequency inverted main-path and non-inverted feedforward signals cancel — origin of the (sCGD−gm) numerator

Zero caused by feedforward path through CGD





Right-Half-Plane Zero — Quantitative Derivation

- From slide 19's exact transfer function, the numerator's (sCGD−gm) factor vanishes at s = gm/CGD
- fz = +gm/(2πCGD) — a Right-Half-Plane (RHP) zero, at s = +gm/CGD rather than −gm/CGD
- An RHP zero boosts magnitude but adds phase lag (like a pole) — troublesome for feedback-amplifier compensation

Right half plane zero, from the numerator of vo/vi

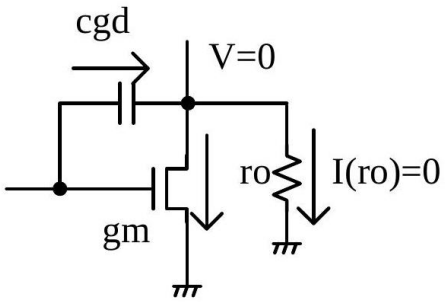
$$\frac{vo}{vi}=\frac{(sC_{GD}-g_m)R_D}{s^2R_SR_D(C_{GS}C_{GD}+C_{GS}C_{SB}+C_{GD}C_{DB})+s\left[R_S(1+g_mR_D)C_{GD}+R_SC_{GS}+R_D(C_{GD}+C_{DB})\right]+1}$$

$$\frac{sC_{GD}-g_m}{\cdots}\quad\rightarrow\quad f_z=\frac{+g_m}{2\pi C_{GD}}$$

Right-Half-Plane Zero — Quick Physical Derivation

- Shortcut: set Vout = 0 (the defining condition for a zero) and solve for s
- With Vout = 0, current through ro is zero, so gm·Vgs must be balanced entirely by current through CGD
- gm·Vgs = sCGD·Vgs ⟹ sZ = gm/CGD — matches slide 23, derived without solving the full transfer function

Right Half Plane Zero



$$g_m V_{gs} = sC_{GD}V_{gs} \Rightarrow Current\ through\ r_o\ is\ zero$$

$$\therefore s_Z = g_m / C_{GD}$$

OCTC Dominant-Pole Result — Summary

- D = (s/ωp1+1)(s/ωp2+1) ≈ 1 + s/ωp1 when ωp2 >> ωp1, dropping the 1/ωp2 and s² terms
- Boxed result (emphasized in original slide): fp,in = 1/{2π(RS[CGS+(1+gmRD)CGD] + RD(CGD+CDB))}
- Key summary of the CS-stage Miller/OCTC analysis, before moving to input impedance

➢**Open-Circuit Time Constant Approximation**

$$Assume\ D=\left(\frac{s}{\omega_{p1}}+1\right)\left(\frac{s}{\omega_{p2}}+1\right)=\frac{s^2}{\omega_{p1}\omega_{p2}}+\left(\frac{s}{\omega_{p1}}+\frac{s}{\omega_{p2}}\right)+1,$$

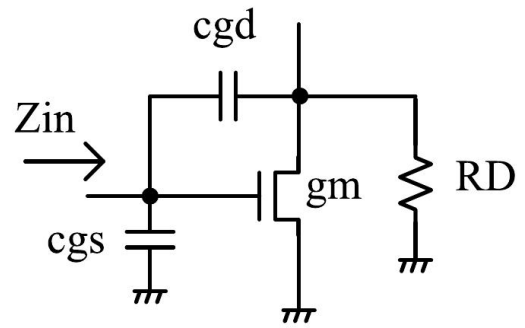
$$\omega_{p2}>>\omega_{p1}, D\approx\frac{s^2}{\omega_{p1}\omega_{p2}}+\left(\frac{s}{\omega_{p1}}\right)+1 \Rightarrow \boxed{D\approx\left(\frac{s}{\omega_{p1}}\right)+1}$$

$$f_{p,in}=\frac{1}{2\pi\left(R_S\left[C_{GS}+(1+g_mR_D)C_{GD}\right]+R_D(C_{GD}+C_{DB})\right)}$$

Common-Source Input Impedance

- Simplified high-frequency model looking into the gate (RS omitted): cgs to ground, cgd feeding back to drain, gm and RD at output
- Zin = 1/{s·[CGS+(1+gmRD)·CGD]} — purely capacitive, a single equivalent capacitor
- Same Miller-multiplied capacitance as the ωp,in denominator — shows why high-gain CS stages present large input capacitance

➢**Input Impedance**



$$Z_{in} = \frac{1}{s\left[C_{GS} + (1 + g_m R_D) C_{GD}\right]}$$

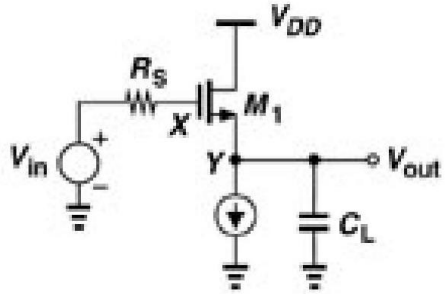


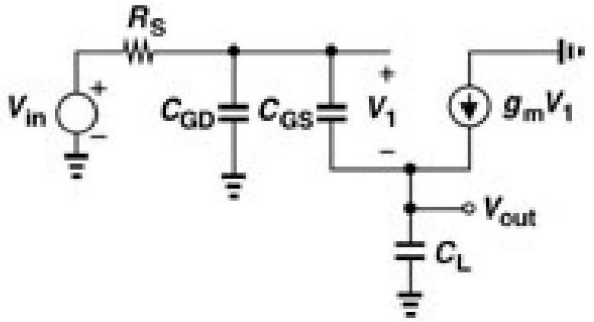
Source Follower — Frequency Response

- Exact transfer function and OCTC dominant pole
- Input impedance at low and high frequency
- Output impedance, negative resistance, and ringing

Source Follower — Exact Transfer Function

- Common-drain stage: M1 drain at VDD, gate driven through RS, source is the output with bias current source and load CL
- vo/vi = (gm+sCGS) / [s²RS(CGSCL+CGSCGD+CGDCL) + s(gmRSCGD+CL+CGS) + gm]
- Zero at s = −gm/CGS is left-half-plane (unlike the CS stage's RHP zero), since there is no inverted feedforward path





$$\frac{vo}{vi} = \frac{g_m + sC_{GS}}{s^2 R_S (C_{GS}C_L + C_{GS}C_{GD} + C_{GD}C_L) + s\left(g_m R_S C_{GD} + C_L + C_{GS}\right) + g_m}$$

$$f_{p1} \approx \frac{g_m}{2\pi(g_m R_S C_{GD} + C_L + C_{GS})}, \text{ assuming } f_{p2} >> f_{p1}$$

$$= \frac{1}{2\pi\left(R_S C_{GD} + \dfrac{C_L + C_{GS}}{g_m}\right)}$$

Source Follower — OCTC Open-Circuit Resistances

- OCTC rule: each capacitor sees a resistance with all other capacitors open; τ = ΣRiCi approximates the dominant pole
- CGD sees RS; CGS and CL both see ≈1/gm — the resistance looking into the source
- τ = RS·CGD + CGS/gm + CL/gm, ω = 1/τ — matches slide 27's fp1 formula exactly

➢**Open Circuit Time Constant Approximation**

$$C_{GD} \Rightarrow R_S$$

$$C_{GS} \Rightarrow 1/g_m \Rightarrow \textit{See next slide!}$$

$$C_L \Rightarrow 1/g_m$$

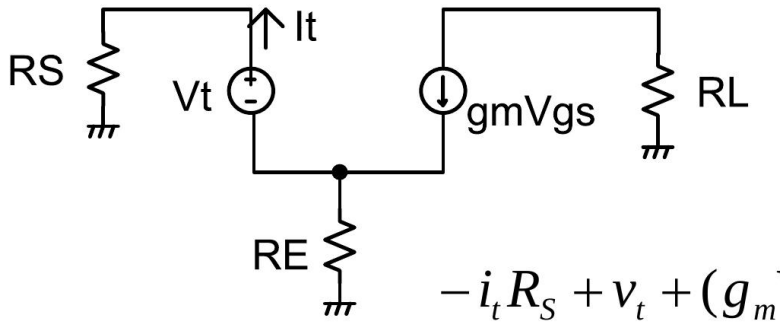
$$\tau = C_{GD}R_S + C_{GS}/g_m + C_L/g_m$$

$$\omega = 1/\tau$$

Why the Source-Looking Resistance Is ≈ 1/gm

- Test circuit: inject it, vt at the source; RS at the gate; general resistance RE also at the source
- Loop/KCL analysis (Vgs = −vt) gives vt/it = (RS+RE)/(1+gmRE)
- Limit RE → ∞ (as used for CGS, CL on slide 28): vt/it → 1/gm — proves the OCTC resistance assignment

➢**Open Circuit Time Constant Approximation**



$$-i_t R_S + v_t + (g_m V_{gs} - i_t) R_E = 0$$

$$-i_t R_S + v_t + (g_m v_t - i_t) R_E = 0$$

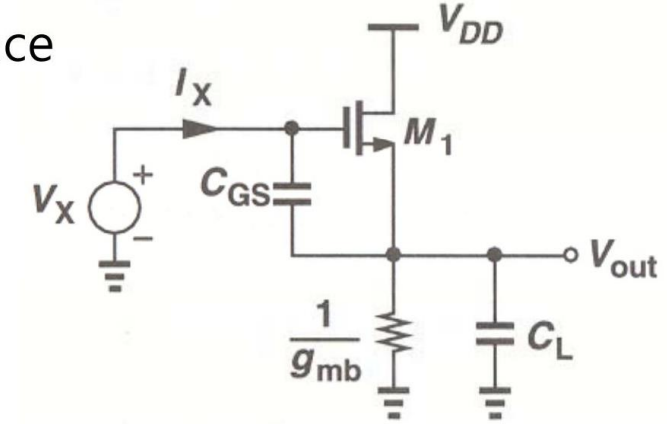
$$\therefore v_t / i_t = \frac{R_S + R_E}{1 + g_m R_E}$$

$$when\ R_E = \infty \Rightarrow R_t = 1/g_m$$

Source Follower — Input Impedance (General)

- Test source VX, IX at the gate; CGS to ground at gate; source node has 1/gmb (body effect) in parallel with load CL
- VX = IX/(sCGS) + (IX + gm·IX/sCGS)·(1/gmb ∥ 1/sCL)
- Neglecting CGD: Zin ≈ 1/sCGS + (1+gm/sCGS)/(gmb+sCL) — examined at low/high frequency limits next

➢Input impedance



$$V_X = I_X / sC_{GS} + (I_X + g_m(I_X / sC_{GS}))(1/g_{mb} \| 1/sC_L)$$

$$\therefore Z_{in} = V_X / I_X = 1/sC_{GS} + (1 + g_m / sC_{GS})(1/g_{mb} \| 1/sC_L)$$

$$\text{Neglecting } C_{GD},\ Z_{in} \approx \frac{1}{sC_{GS}} + \left(1 + \frac{g_m}{sC_{GS}}\right)\frac{1}{g_{mb} + sC_L}$$

Source Follower Zin — Low-Frequency Limit

- gmb >> |sCL|: (1/gmb∥1/sCL) ≈ 1/gmb, so Zin ≈ (1/sCGS)(1+gm/gmb) + 1/gmb
- Cin = CGS·[1−gm/(gm+gmb)] + CGD — exactly the Miller formula C(1−Av) with Av ≈ gm/(gm+gmb)
- Since Av ≈ +1, the Miller factor ≈ 0: CGS shrinks instead of growing — source followers have low, wideband input capacitance

➢Input impedance

At low frequencies, $g_{mb} >> |sC_L|$

$$Z_{in} \approx \frac{1}{sC_{GS}}\left(1+g_m/g_{mb}\right)+1/g_{mb}$$

$$\therefore C_{in} = C_{GS}\left[1-g_m/\left(g_m+g_{mb}\right)\right]+C_{GD}$$

$$C\times(1-A_v) = \text{Miller capacitance}$$

$$= C_{GS}g_{mb}/(g_m+g_{mb})+C_{GD} \quad (\textit{same as Miller})$$

Source Follower Zin — High-Frequency Limit

- gmb << |sCL|: Zin ≈ 1/sCGS + 1/sCL + gm/(s²CGSCL)
- First two terms: CGS and CL in series; third term = −gm/(ω²CGSCL), a NEGATIVE resistance shrinking with frequency²
- Zin = CGD ‖ (CGS series CL) plus this negative resistance — root cause of possible instability with reactive loads

Neglecting C_{GD}, $Z_{in} \approx \frac{1}{sC_{GS}} + \left(1 + \frac{g_m}{sC_{GS}}\right)\frac{1}{g_{mb} + sC_L}$

At high frequencies, $g_{mb} << |sC_L|$

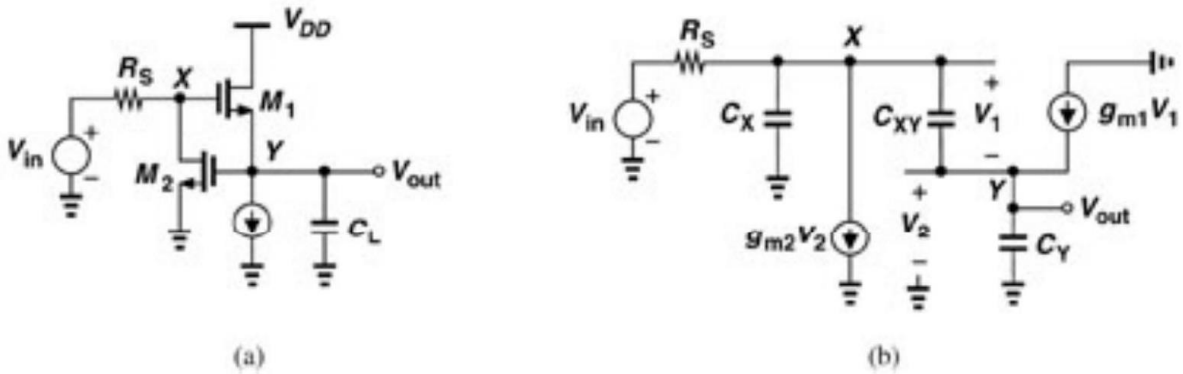
$$Z_{in} \approx \frac{1}{sC_{GS}} + \frac{1}{sC_L} + \frac{g_m}{s^2 C_{GS} C_L}$$

At a particular frequency, input impedance includes C_{GD} in parallel with series combination of C_{GS} and C_L and a *negative* resistance equal to $-g_m/(C_{GS}C_L\omega^2)$.

Ex. 6.6 — Two-Stage Source Follower (Left as Exercise)

- Two cascaded source followers M1, M2 (M2 as active load) with CX, CXY (feedback), and CY
- Instructor assigns this as self-study: 'Please solve this by yourself... we are time-pressed!'
- Structurally an interacting-poles problem (like slide 19) with an extra feedback capacitor CXY

Ex6.6) calculate the transfer function



Please, solve this by yourself.
Usually, I have not skipped the examples.
But, this time we are time-pressed!!!

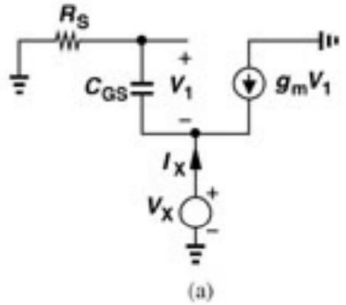
Source Follower Output Impedance — Algebraic

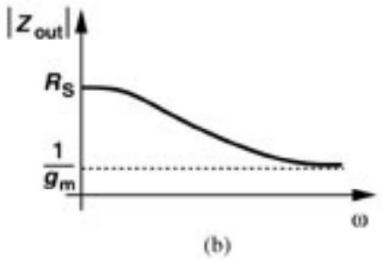
- Test source VX, IX at the source; RS at gate; CGS between gate/source; dependent source gm·V1
- Solving: Zout = VX/IX = (CGS·s·RS+1)/(gm+CGS·s) — a one-pole, one-zero impedance
- Highlighted: case (c), |Zout| rising with frequency from 1/gm toward RS, suggests an inductive component in Zout

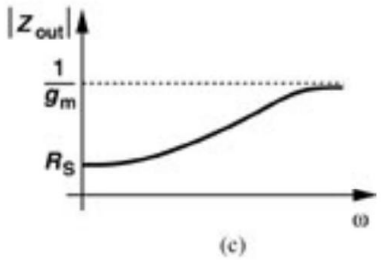
$$V_1 C_{GS} s + g_m V_1 = -I_x, V_1 C_{GS} s R_s + V_1 = -V_x$$

$$Z_{out} = V_x / I_x = \frac{C_{GS} s R_s + 1}{g_m + C_{GS} s}$$

(c) is more likely that we can consider inductive component in Zout

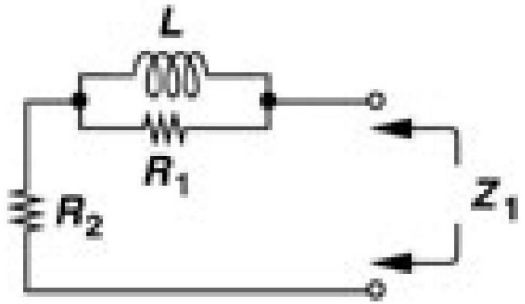




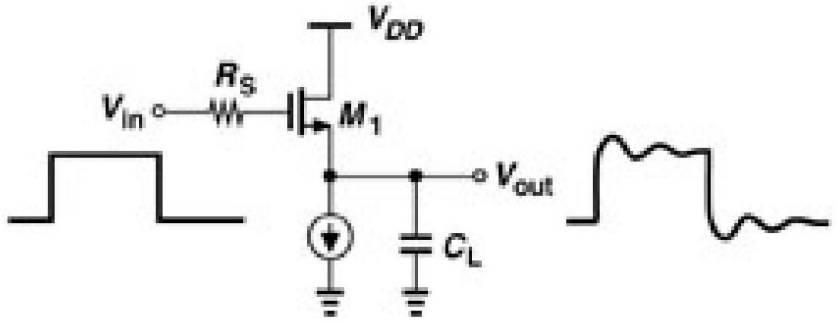


Ringing in the Source-Follower Step Response

- Equivalent Zout modeled as RLC: inductor L ‖ R1 (captures rising impedance), whole branch ‖ R2
- Driving a heavy capacitive load CL with a voltage step produces visible RINGING — underdamped (Q>0.5) response
- Design warning: a bare source follower buffering a large CL can excite ringing/overshoot without added damping



Equivalent output impedance of a source follower



Ringing in step response of a source follower with a heavy capacitive load

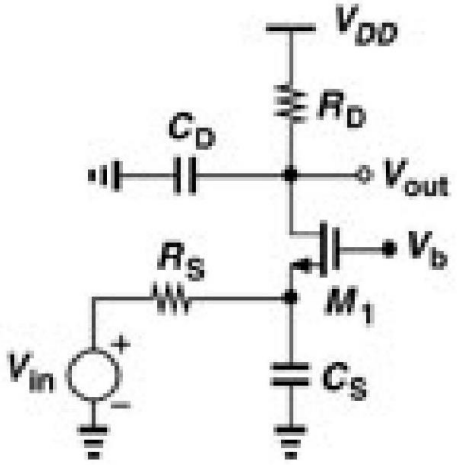
VI

Common-Gate Stage — Frequency Response

- No Miller capacitance — wider bandwidth than CS
- Ex. 6.7: input impedance independent of CL

Common-Gate Stage — No Miller Capacitance

- CG topology: gate at AC ground (Vb), signal in at source (via RS), signal out at drain (RD to VDD)
- Highlighted: 'No Miller capacitance here' — no capacitor bridges input/output with large inverting gain
- DC gain Av = (gm+gmb)RD/[1+(gm+gmb)RS]; poles from CS (input) and RD·CS (output) — both higher frequency than an equivalent CS stage



No Miller capacitance here

Input impedance seen looking into the source of M1

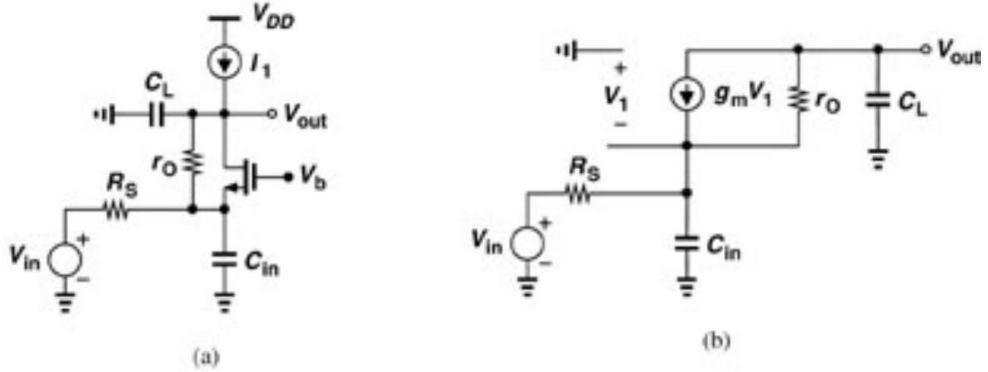
$$Z_{in} \approx \frac{Z_L}{(g_m+g_{mb})r_o} + \frac{1}{(g_m+g_{mb})}$$

$$A_v = \frac{(g_m+g_{mb})R_D}{1+(g_m+g_{mb})R_S} \frac{1}{\left(1+\frac{C_S}{g_m+g_{mb}+R_S^{-1}}\right)(1+R_D C_S s)}$$

Ex. 6.7 — Why CG Zin Is Independent of CL

- Zin = 1/(gm+gmb) + [1/(CL·s)]·[1/((gm+gmb)·ro)]
- The CL-dependent term is attenuated by the transistor's large intrinsic gain (gm+gmb)ro
- As CL increases, this attenuated term vanishes: Zin → 1/(gm+gmb), regardless of CL — good for cascode/LNA input stages

Ex6.7) explain why Zin becomes independent of CL as this capacitance increases



$$Z_{in} = \frac{1}{g_m + g_{mb}} + \frac{1}{C_L s} \cdot \frac{1}{(g_m + g_{mb}) r_o}$$

as C_L increases, Z_{in} approaches $1/(g_m + g_{mb})$

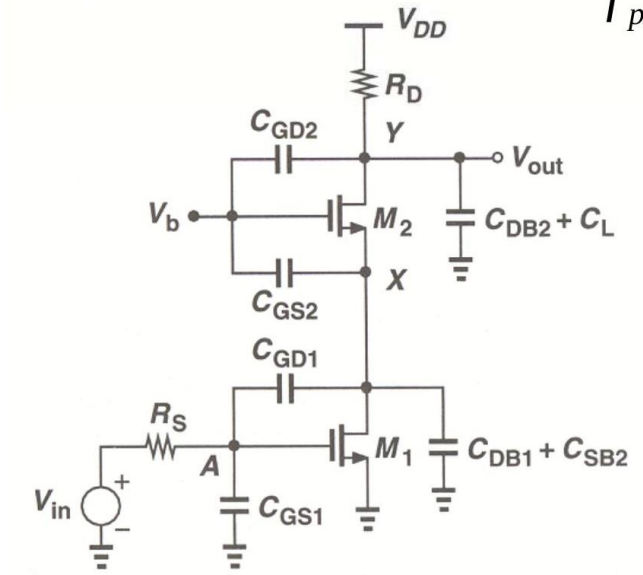
VII

Cascode Stage — Frequency Response

- Pole at each node; suppressed Miller effect at the input
- Gain, input/output resistance vs. plain CS
- Ex. 6.8: transimpedance transfer function

Cascode Stage — Pole at Each Node (A, X, Y)

- CS stage M1 stacked with CG stage M2: node A (gate M1), node X (drain M1/source M2), node Y (drain M2 = output)
- fpA = 1/{2πRS·[CGS1+CGD1·(1+gm1/(gm2+gmb2))]} — small Miller factor, node X has low impedance ≈1/(gm2+gmb2)
- fpX = (gm2+gmb2)/[2π(CGD1+CDB1+CSB2+CGS2)]; fpY = 1/[2πRD(CDB2+CL+CGD2)] — same form as a plain CS output pole



$$f_{pA} = \frac{1}{2\pi R_S\left[C_{GS1} + C_{GD1}\left(1 + \frac{g_{m1}}{g_{m2} + g_{mb2}}\right)\right]}$$

$$f_{pX} = \frac{g_{m2} + g_{mb2}}{2\pi\left(C_{GD1} + C_{DB1} + C_{SB2} + C_{GS2}\right)}$$

$$f_{pY} = \frac{1}{2\pi R_D\left(C_{DB2} + C_L + C_{GD2}\right)}$$

Cascode vs. Common-Source — Three Key Properties

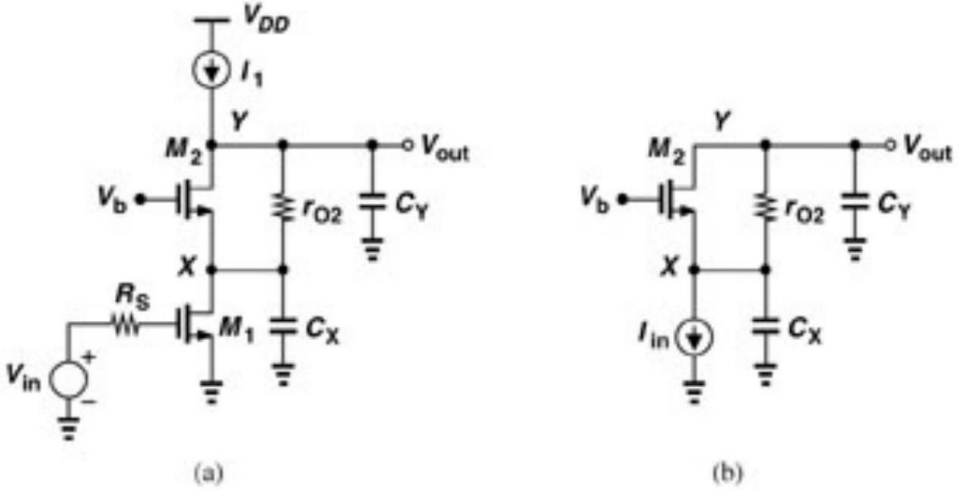
- 'No Miller Effect ⟹ Higher 3-dB Frequency': small gain A→X keeps CGD1's Miller factor minimal, widening bandwidth
- 'Voltage gain not larger than common source': with the same RD, DC gain is still set mainly by gm1·RD
- 'Input/Output resistance equal to the CS amplifier' — with resistive RD, terminal resistances are essentially unchanged

- ➢ No Miller Effect => Higher 3-DB Frequency
- ➢ Voltage gain not larger than common source
- ➢ Input and Output resistance equal to the common source amplifier

Ex. 6.8 (Setup) — Cascode as a Transimpedance Amplifier

- Current-driven cascode: Iin injected at node X, isolating the M2/ro2/CX/CY sub-circuit for analysis
- Vx = −(Vout·CY·s+Iin)/(CX·s); Iro2 = −Vout·CY·s − gm2·(Vout·CY·s+Iin)/(CX·s)
- Vout = Vx + Iro2·ro2 — these three equations are solved on the next slide

Ex6.8) calculate the transfer function



$$V_x = -(V_{out}C_y s + I_{in})/(C_x s)$$

$$I_{r_{o2}} = -V_{out}C_y s - g_{m2}(V_{out}C_y s + I_{in})/(C_x s)$$

$$V_{out} = V_x + I_{r_{o2}} \cdot r_{o2}$$

Ex. 6.8 (Result) — Cascode Transimpedance and Voltage Gain

- Vout/Iin = −(gm2ro2+1)/(CX·s) · 1/{1+(1+gm2ro2)(CY/CX)+CY·ro2·s} — single-pole (M1 idealized as a current source)
- Combined: Vout/Vin = −gm1gm2/(CY·CX·s) · 1/(gm2/CX+s)
- 'Two poles: one is gm/Cx, the other is at origin' — the origin pole is an artifact of no finite ro1 limiting DC gain

Ex6.8) calculate the transfer function

$$\frac{V_{out}}{I_{in}} = -\frac{g_{m2}r_{o2}+1}{C_X s}\cdot\frac{1}{1+(1+g_{m2}r_{o2})\dfrac{C_Y}{C_X}+C_Y r_{o2}s}$$

$$\frac{V_{out}}{V_{in}} = -\frac{g_{m1}g_{m2}}{C_Y C_X s}\cdot\frac{1}{g_{m2}/C_X+s}$$

two poles: one is gm/Cx, the other is at origin

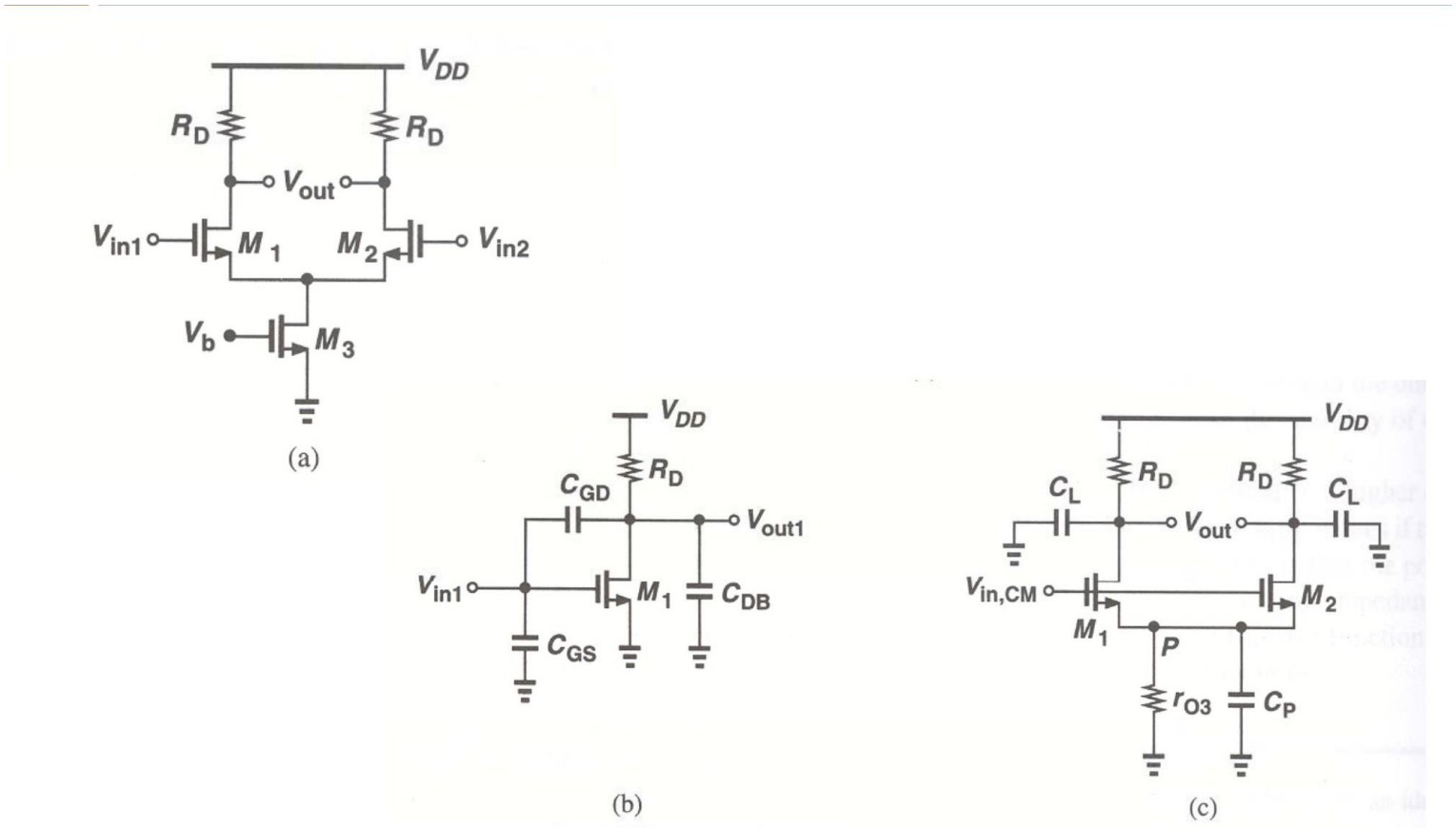
VIII

Differential Pair — Frequency Response

- Differential/common-mode half-circuits and CMRR vs. frequency
- 5-transistor current-mirror-load OTA: mirror pole, output pole, zero
- Fully differential pair: no mirror pole

Differential Pair — Full Circuit and Half-Circuit Models

- (a) M1, M2 with RD loads, tail current source M3 (bias Vb)
- (b) Differential half-circuit: tail node is a virtual ground under DM excitation, reducing to a plain CS stage — same as slides 17-26
- (c) Common-mode half-circuit: tail node P carries ro3 and CP, setting high-frequency CMRR behavior (next slides)

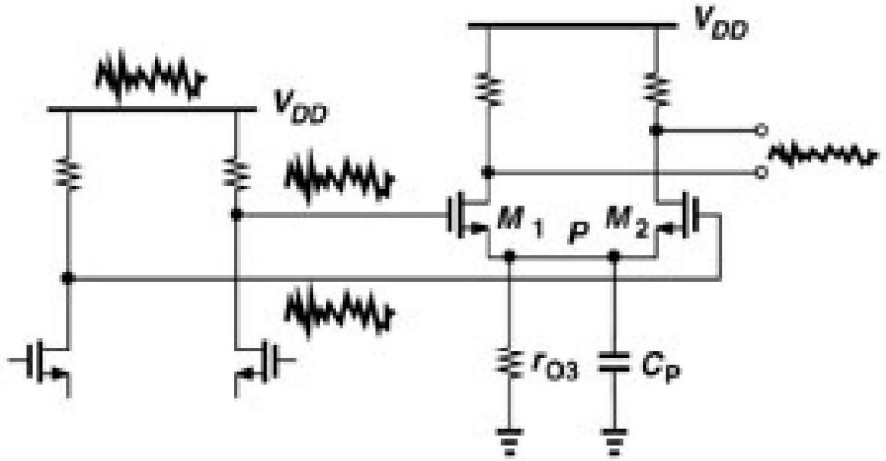


Common-Mode-to-Differential-Mode Gain (Av,CM-DM)

- Av,CM−DM = −Δgm·[RD∥1/(CLs)] / {(gm1+gm2)·[ro3∥1/(CPs)]+1}, with Δgm = gm1−gm2 (device mismatch)
- CP = CGD3+CDB3+CSB1+CSB2 — total parasitic capacitance at tail node P
- Δgm ≠ 0 lets common-mode tail-current variation leak unevenly into the outputs; CP∥ro3 makes this worse at high frequency

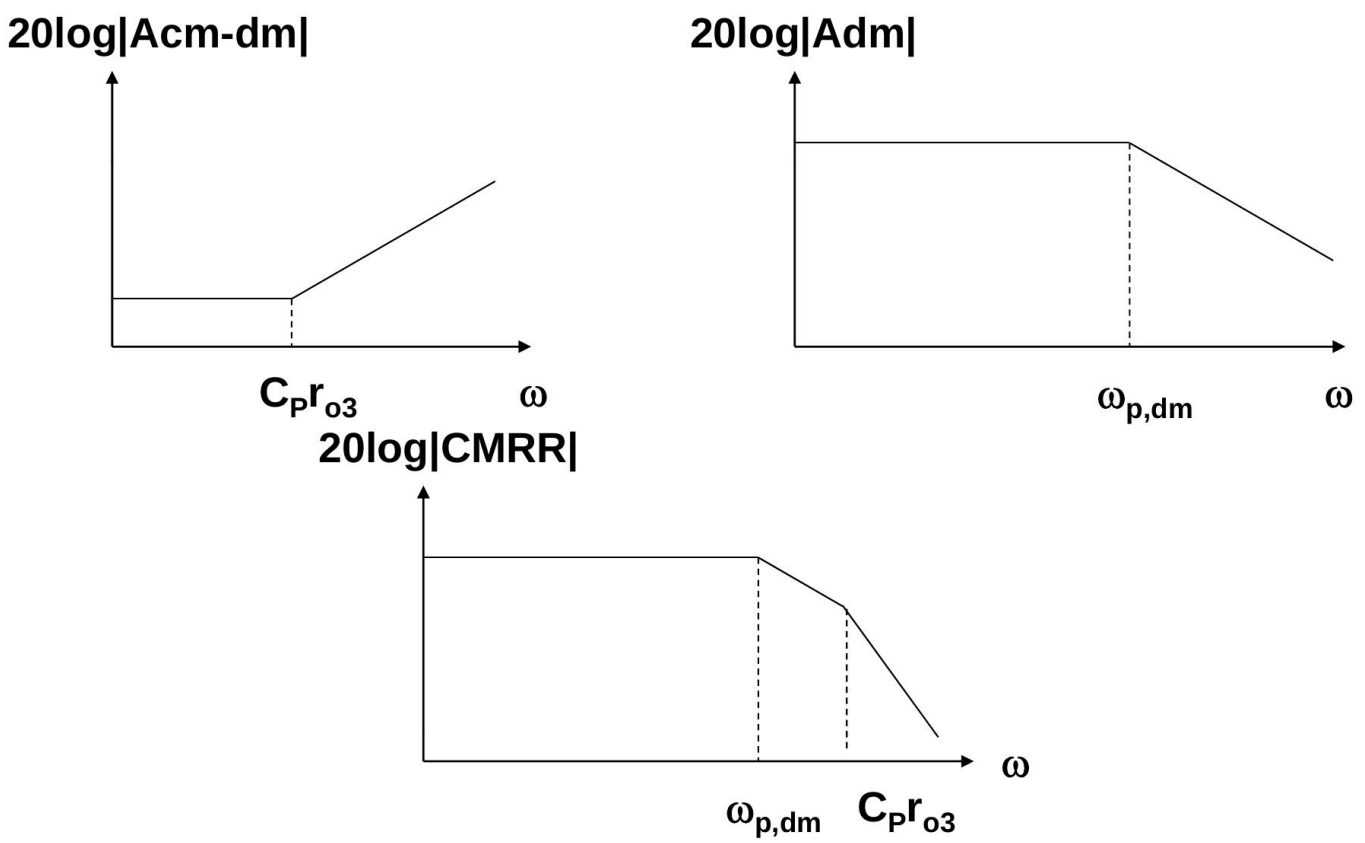
$$A_{v,CM-DM} = -\frac{\Delta g_m \left[R_D \parallel \frac{1}{C_L s} \right]}{(g_{m1} + g_{m2}) \left[r_{o3} \parallel \frac{1}{C_P s} \right] + 1}$$

$$where\, C_P = C_{GD3} + C_{DB3} + C_{SB1} + C_{SB2}$$



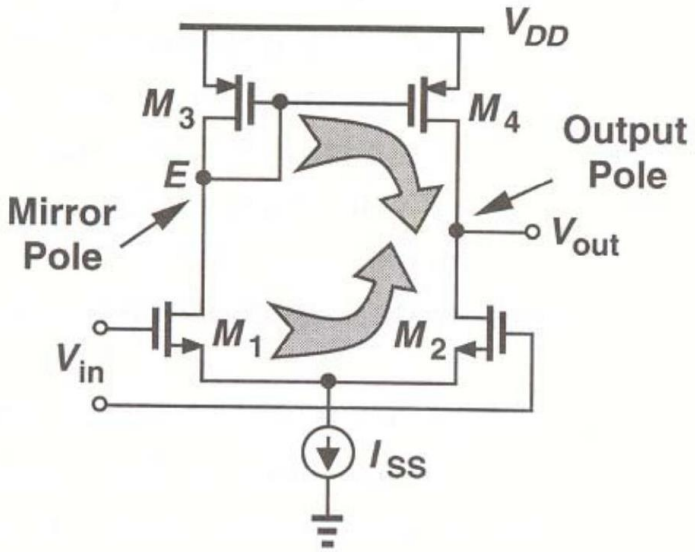
CMRR Degradation with Frequency — Bode Plots

- |Acm-dm| flat at low frequency, then RISES past break frequency ω = CP·ro3 (the tail-node pole)
- |Adm| flat at DC gain, then falls at −20 dB/dec past ωp,dm (the differential half-circuit's dominant pole)
- CMRR = Adm/Acm-dm falls at ωp,dm and falls faster still past CP·ro3 — CMRR always degrades at high frequency



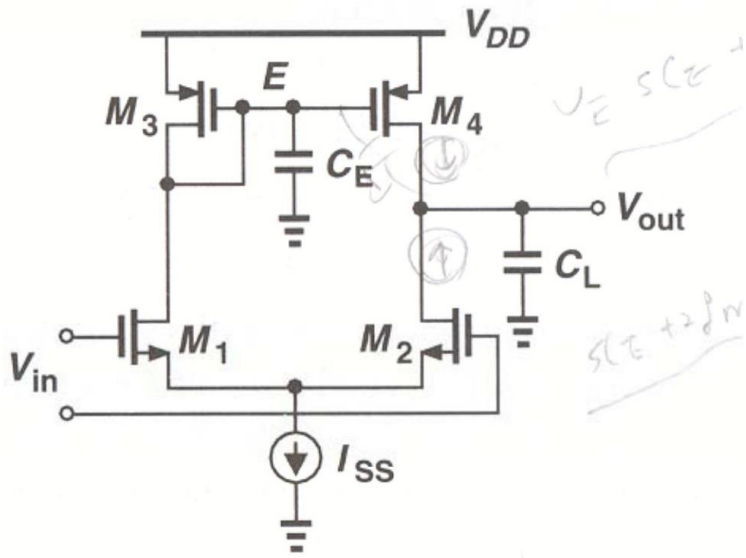
5-Transistor OTA — Current-Mirror Load Topology

- M1, M2 input differential pair; M3 (diode-connected, node E) and M4 form a PMOS current-mirror load for single-ended output
- Signal from M1 flows through the mirror (M3→M4) to node E, the 'Mirror Pole'; M2's signal goes directly to Vout, the 'Output Pole'
- These differently-delayed paths are the physical origin of two poles plus a zero, analyzed next



Mirror-Node Capacitance CE

- CE = CGS3+CGS4+CDB3+CDB1+CGD1+CGD4·(1−Av4) — all capacitances tied to node E
- CGS3, CGS4, CDB3, CDB1, CGD1 all add directly since their terminals land at E
- CGD4·(1−Av4) is Miller-multiplied (M4 acts like a second CS stage) — usually the dominant term, making the mirror pole relatively low



$$C_E = C_{GS3} + C_{GS4} + C_{DB3} + C_{DB1} + C_{GD1} + C_{GD4}(1 - A_{v4})$$

5-Transistor OTA — Three Key Pole/Zero Frequencies

- fp1 ≈ 1/[2π(roN∥roP)CL] — output pole, usually dominant since roN∥roP is large
- fp2 = gm3/(2πCE) — mirror pole, using diode-connected M3's 1/gm3 and CE from the previous slide
- fZ = 2·fp2 = 2gmP/(2πCE) — a zero at exactly twice the mirror-pole frequency (derivation on next slide)

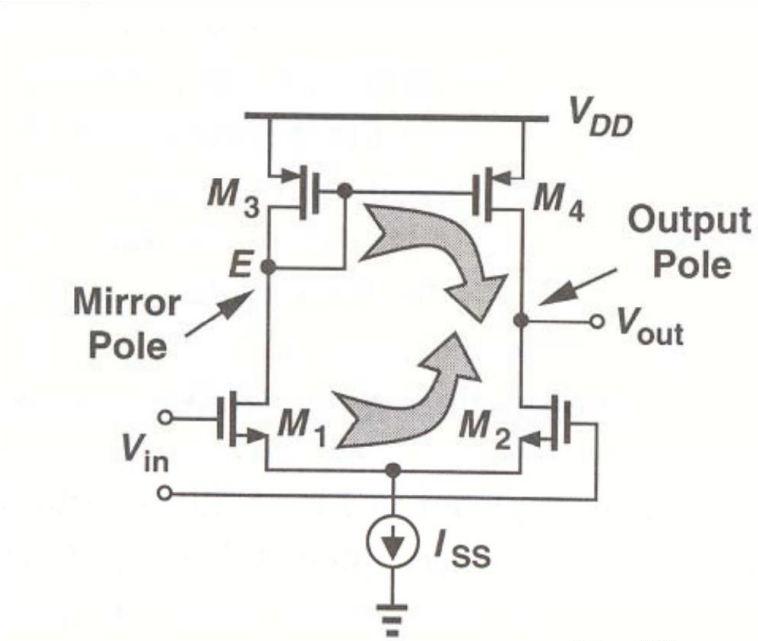
$$f_{p1} \approx \frac{1}{2\pi(r_{oN} \parallel r_{oP})C_L}$$

$$f_{p2} = \frac{g_{m3}}{2\pi C_E}$$

$$f_Z = 2 f_{p2} = \frac{2g_{mP}}{2\pi C_E} \Rightarrow \textit{See next slide!}$$

5-Transistor OTA — Complete Transfer Function and Zero

- Vout/Vin = A0/(1+s/ωp1) · [1/(1+s/ωp2)+1] — M2's direct path (gain 1) plus M1's mirror-delayed path (1/(1+s/ωp2))
- Simplified: A0(2+s/ωp2)/[(1+s/ωp1)(1+s/ωp2)]; numerator zero at s = −2ωp2, i.e. ωz = −2gmP/CE
- Left-half-plane zero (unlike the CS stage's RHP zero) sitting close to fp2 — partly compensates fp2's rolloff



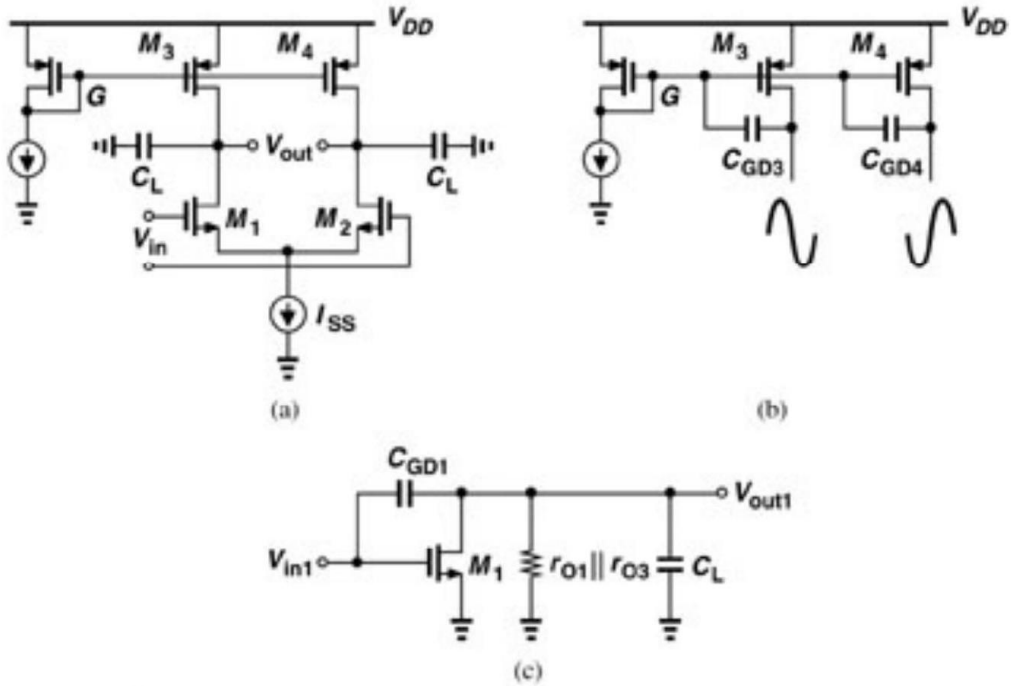
$$V_{out}/V_{in} = \frac{A_0}{1+s/\omega_{p1}}\left(\frac{1}{1+s/\omega_{p2}}+1\right) = \frac{A_0}{1+s/\omega_{p1}}\cdot\frac{2+s/\omega_{p2}}{1+s/\omega_{p2}}$$

$$Zero\ at\ s = -2\omega_{p2}$$

$$\therefore \omega_z = -2g_{mp}/C_E$$

Fully Differential Pair — No Mirror Pole

- Separate PMOS mirror devices M3, M4 (no shared diode-connected node), each branch has its own output with load CL
- Each branch produces its own differential output directly — no need to mirror signal across branches
- Half-circuit is a plain CS stage; conclusion: 'No mirror pole', 'Output pole is dominant' — simpler than the 5-T OTA



No mirror pole
Output pole is dominant